

QIAN MA

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2027 Full-Time internship availability: Spring + Summer or Summer + Fall in US; Available for a China-based role in Summer 2027; U.S. F-1 visa valid through November 2028.

🎓 EDUCATION

Rensselaer Polytechnic Institute Jan. 2024 – Jan. 2028 (expected)

Ph.D. Student in Computer Science, advised by [Prof. Yao Ma](#)

City University of Hong Kong 2022 – 2024

M.Sc. in Multimedia Information Technology, Distinction

University of Electronic Science and Technology of China 2018 – 2022

B.Eng. in Software Engineering - Systems and Technology

⚙️ RESEARCH INTERESTS AND EXPERTISE

- Multimodal AI: computer vision, vision-language models (VLMs/MLLMs), multimodal RAG, knowledge-based VQA, visual entity grounding, reliable inference, and model abstention.
- Methods and prior expertise: activation patching, representation steering, foundation-model evaluation, graph learning, graph self-supervised learning, and spatio-temporal modeling.

👤 RESEARCH EXPERIENCE

DAMI-Lab, Rensselaer Polytechnic Institute Jan. 2024 – Present

Ph.D. Student Researcher Advisor: [Prof. Yao Ma](#)

- Lead first-author projects on multimodal RAG and KB-VQA, studying how VLM/MLLM systems ground visual queries to external knowledge [1][2].
- First-authored ACL 2026 work [2] showing that KB-VQA benefits from separating visual entity identification from evidence selection instead of entangling retrieval, grounding, and ranking.
- First-authored ECCV 2026 work [1] auditing KB-VQA benchmarks for unsupported answers, underspecified questions, and visually simplified single-entity settings.
- First-authored graph foundation-model work on graph SSL scaling/evaluation [3][4] and LLM-guided graph structure learning [5].

AML Lab, City University of Hong Kong Oct. 2022 – Jun. 2023

Master Student Research Assistant Mentor: [Zijian Zhang](#); Advisors: [Prof. Xiangyu Zhao](#), [Prof. Haoliang Li](#)

- First-authored HIRST [6] (CIKM 2023), modeling sensor dependencies through regional and global hierarchies for traffic forecasting; the project received an A+ for the graduate dissertation.
- Collaborated on *PromptST* [7] (CIKM 2023), covering prompt learning and multi-attribute time-series prediction for smart-city applications.

University of Electronic Science and Technology of China Dec. 2019 – Jun. 2020

Undergraduate Research Assistant Supervisor: Prof. Su Sheng

- Processed large-scale urban driving records and implemented MapReduce-style parallel preprocessing for grid-based urban computing research.

INDUSTRY RESEARCH AND ENGINEERING EXPERIENCE

AT&T Chief Data Office, Bedminster, NJ

Jun. 2026 – Aug. 2026

Graduate Research Intern Mentors: [Qiong Wu](#), [Sanjeev Misra](#); Manager: [Emily M. Dodwell](#)

- Investigating whether MLLMs can leverage their internal knowledge and latent verification mechanisms to abstain when retrieved evidence is insufficient.
- Developing inference-time methods based on activation patching and representation steering to strengthen abstention while preserving correct visual entity identification (ongoing work).

AT&T Chief Data Office, Bedminster, NJ

May 2025 – Aug. 2025

Graduate Research Intern Mentors: [Qiong Wu](#), [Zhengyi Zhou](#); Manager: [Divesh Srivastava](#)

- Conducted research on multimodal RAG for knowledge-grounded visual question answering, focusing on identifying visual entities before retrieving and reasoning over external evidence.
- Built reproducible datasets, prompts, evaluation code, and retrieval baselines; the research led to first-authored ECCV 2026 and ACL 2026 papers [\[1\]\[2\]](#).

SAP Labs China, Xi'an

Apr. 2021 – May 2022

Software Engineering Intern, SAP Data Intelligence Manager: [Jingtao Li](#)

- Developed and maintained two CI/CD automation microservices for a cloud infrastructure service automation platform and presented projects/features in internal global sharing sessions.

PUBLICATIONS

- [1] **Qian Ma**, S M Rayeed, Charles V. Stewart, Qiong Wu, Yao Ma. *Identifying and Resolving Pitfalls of Knowledge-Based VQA Benchmarks: Auditing, Repairing, and Augmenting*. ECCV 2026, to appear.
- [2] **Qian Ma**, Qiong Wu, Zhengyi Zhou, Yao Ma. *Ground Then Rank: Revisiting Knowledge-Based VQA with Training-Free Entity Identification*. ACL 2026 Findings.
- [3] **Qian Ma**, Haitao Mao, Jingzhe Liu, Zhehua Zhang, Chunlin Feng, Yu Song, Yihan Shao, Yao Ma. *Do Neural Scaling Laws Exist on Graph Self-Supervised Learning?* LoG 2024.
- [4] **Qian Ma**, Hongliang Chi, Hengrui Zhang, Kay Liu, Zhiwei Zhang, Lu Cheng, Suhang Wang, Philip S. Yu, Yao Ma. *Overcoming Pitfalls in Graph Contrastive Learning Evaluation: Toward Comprehensive Benchmarks*. ACM SIGKDD Explorations, 2025.
- [5] **Qian Ma**, Haitao Mao, Zhehua Zhang, Qiong Wu, Zhengyi Zhou, Yao Ma. *Cross-Domain GraphWalker: Harnessing LLMs for Graph Structure Learning*. Under review.
- [6] **Qian Ma**, Zijian Zhang, Xiangyu Zhao, Haoliang Li, Hongwei Zhao, Yiqi Wang, Zitao Liu, Wanyu Wang. *Rethinking Sensors Modeling: Hierarchical Information Enhanced Traffic Forecasting*. CIKM 2023.
- [7] Zijian Zhang, Xiangyu Zhao, Qidong Liu, Chunxu Zhang, **Qian Ma**, et al. *PromptST: Prompt-Enhanced Spatio-Temporal Multi-Attribute Prediction*. CIKM 2023.

TECHNICAL SKILLS

- Multimodal AI: VLMs/MLLMs, multimodal RAG, KB-VQA, visual entity grounding, abstention, activation patching, representation steering, Qwen-VL, LLaVA, Llama, CLIP, FAISS, and BGE reranking.
- Programming and ML: Python, Java, JavaScript, Go, C; PyTorch, PyTorch Lightning, and prompt-based evaluation.
- Systems and Tools: Linux, Git, Docker, Kubernetes, Slurm, Singularity/Apptainer, Jenkins, and $\LaTeX$.
- Languages: Mandarin - native; English - fluent.

HONORS AND AWARDS

KDD 2026 Best Reviewer - Research Track

2026

KDD 2026 Best Reviewer - Datasets & Benchmarks Track

2026

Outstanding Student Scholarship (three academic years)

2018 – 2022